

SimVille: Exploring Social Complexity and Emotional Tone in LLM-Driven Virtual Reality Conversation

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Figure 1: Teaser.

ABSTRACT

We present SimVille, a mixed reality game that combines extended reality (XR), large language models (LLMs), and conversational non-player characters (NPCs) to explore new forms of immersive, socially interactive learning. In SimVille, players engage in quests by conversing naturally with embodied NPCs, where dialogue is dynamically generated through LLMs. The game leverages XR to situate interactions in a shared physical-digital environment, enabling players to move, explore, and connect with NPCs in contextually rich scenarios. Beyond quest completion, the system investigates how adaptive dialogue, social cues, and narrative-driven embodiment can support the development of communication skills, such as small talk, irony detection, or emotion recognition. We describe the design of SimVille, including technical integration of XR and LLM pipelines, interaction mechanics, and a framework for evaluating conversational experiences. Our contribution highlights opportunities and challenges in creating explainable, personalized, and engaging human–AI interactions in augmented and mixed reality.

Index Terms: Game, Large Language Models, Conversation, Extended Reality, Virtual Reality, Game Complexity

*e-mail: blinded@blinded.com

†e-mail: blinded@blinded.com

1 INTRODUCTION

Recent advances in extended reality (XR) and large language models (LLMs) are transforming how humans interact with artificial agents. As conversational models become capable of generating contextually appropriate, affectively aligned dialogue, they enable novel forms of social simulation and training in immersive environments. Virtual agents powered by LLMs can respond dynamically, exhibit empathy, and adapt their communication style to user behavior—opening new opportunities for education, therapy, and social skills development.

Within this context, SimVille explores how naturalistic, emotionally expressive virtual agents can support conversational learning in a socially rich XR setting. The system combines immersive virtual reality (VR) with adaptive LLM-driven dialogue to simulate realistic social encounters involving both cooperative and challenging interactions. Such experiences are particularly relevant for users who may struggle with social communication, including individuals on the autism spectrum, as prior studies have shown that immersive environments can effectively enhance social understanding and emotion regulation when properly tailored to cognitive and sensory needs [13, 9].

Virtual Reality (VR) provides an immersive and controlled environment that allows users to safely experience and practice real-world social interactions. In *SimVille*, VR enables participants, particularly those with social communication challenges to engage in realistic conversations, perceive emotional cues, and receive adaptive feedback without fear of negative judgment. The immersive nature of VR enhances presence and emotional engagement, while maintaining full experimental control over context, tone, and diffi-

culty. This makes VR an ideal tool for studying and training social behavior in a reproducible yet naturalistic way.

Virtual Reality (VR) provides an immersive, controllable, and safe environment for studying and training social communication. In the context of *SimVille*, VR allows participants to engage in realistic conversations with virtual agents while maintaining full experimental control over social complexity, emotional tone, and environmental factors. This immersion enhances presence and emotional engagement, enabling users to perceive nonverbal cues and practice adaptive behaviors without fear of social judgment. Moreover, the reproducibility of virtual scenarios supports systematic evaluation of conversational performance, making VR an ideal medium for investigating how adaptive, LLM-driven dialogue can foster social understanding and emotional regulation.

The study investigates how the number of conversational partners and their emotional tone influence user engagement, comfort, and cognitive load in a virtual restaurant scenario. Two research questions guide the experiment:

1. RQ1: How does the number of conversational partners in the scene affect user engagement and comfort?
2. RQ2: How does the emotional tone of NPCs influence user experience and cognitive load?

2 RELATED WORK

Ongoing research investigates the use of large language models (LLMs) in diverse domains, including arithmetic, common sense reasoning, and symbolic processing [12], as well as linguistics, understanding of the physical world, temporal reasoning, and social intelligence [4]. Beyond information retrieval and conversational support, LLMs can also be guided to assume specific roles. This approach, known as role prompting within prompt engineering, shapes model output to better align with desired objectives. For instance, it enables LLMs to generate contextually accurate responses when queried about historical events or other specialized areas of knowledge [6].

LLMs demonstrate considerable potential for simulating aspects of daily human life. Evaluation studies show that generative agents [19] are capable of producing individual and emergent social behaviors through structured phases of planning, scoring, and reflection. These phases serve as essential components for recalling past interactions and enabling proactive agents that can initiate contextually appropriate actions based on user input.

In gaming contexts, LLMs have demonstrated the ability to simulate human life by reproducing interactions among individuals within a virtual village [19]. Beyond entertainment, Artificial Intelligence (AI) agents, such as chatbots and virtual assistants, are increasingly being applied in healthcare care, where they can provide support and disseminate information using specialized training data [11]. Accurate representations of virtual humans are particularly important for training social skills in immersive environments, as they allow users to develop a deeper sense of empathy, especially when the virtual character is able to provide empathic feedback [5].

The use of virtual avatars in assistive learning has gained increasing attention in recent research. Systematic reviews highlight substantial benefits for learners, including improvements in social skills, emotional competencies, daily living abilities, communication, and attention through interactions with avatars [14, 16]. Gamified online role-play has also shown promise; for example, a tele-dentistry study reported improved self-perceived confidence and awareness when a professional actress was trained to play a patient with a dental condition [24]. Furthermore, experimental studies demonstrate that the integration of empathic modules - capable of simulating emotional states and offering clarifications - significantly enhances perceived empathic listening and emotional intelligence, thus strengthening learner trust [1].

The potential benefits of Large Language Models (LLMs) span a wide range of applications, including question answering, decision support, information retrieval, and multimodal medical tasks [18]. In healthcare, LLMs are particularly promising due to their ability to follow instructions, perform advanced natural language processing, and adapt to contextual information. Their integration into clinical workflows has the potential to enhance decision-making, improve patient care, and reduce medical errors [11]. Furthermore, LLMs have demonstrated the ability to simulate reasoning of the mental state, achieving performance comparable to humans in Theory of Mind (ToM) assessments [23, 22, 3, 20].

Previous research has demonstrated that virtual reality (VR) can serve as an effective medium for enhancing social skills in individuals with autism spectrum disorder (ASD) [13]. However, the intensity and complexity of sensory stimuli within VR environments necessitate careful adaptation to the individual's perceptual and cognitive needs [9]. Consequently, it is essential to design VR experiences that account for varying levels of difficulty, ensuring that each participant remains within an optimal range of motivation and engagement without experiencing excessive stress or sensory overload.

Recent research highlights both the promise and the current limitations of using extended reality (XR) technologies to support the development of social skills in autistic people. Systematically reviewed studies using XR for social skills training reported significant improvements in 37% of the interventions examined [17]. The highest success rates were achieved with augmented reality (AR; 80%), followed by immersive virtual reality (VR; 50%) and non-immersive VR (23%). The comparatively lower effectiveness of VR was attributed to usability challenges and sensory difficulties experienced by participants. The authors emphasized the importance of structured training for both therapists and participants to enhance intervention outcomes. XR-based interventions [7] in multiple developmental domains, including social interaction, recognition and regulation of emotions, communication, attention, and reduction of anxiety. Among the studies on emotion recognition and control, 14 were identified, ten of which used AR applications on mobile devices. These systems overlaid virtual cues such as textual hints or animated facial expressions onto real-world contexts to help participants identify fundamental emotions, including happiness, sadness, fear, and anger.

A virtual reality-based framework was developed to assess emotional everyday scenarios with varying levels of social complexity as a foundation for future digital health applications targeting individuals with autism spectrum disorder (ASD). The study combined subjective ratings and psychophysiological measures (ECG, EDA) to validate that immersive environments can differentiate emotional and safety responses across conditions, demonstrating the therapeutic potential of VR for social and emotional training [26].

3 METHOD

The prototype system was developed in Unity using the XR Interaction Toolkit to create immersive virtual reality environments and manage user interactions on the Meta Quest 3 headset. The framework ran on a workstation equipped with an NVIDIA RTX 4070 Ti GPU, an Intel Core i7-14700K processor, and 32 GB of RAM. Avatars were implemented using Ready Player Me characters, which were imported into Unity with integrated rigging, facial blendshapes, and animation compatibility. This approach allowed for the creation of diverse, realistic, and easily customizable virtual agents that support consistent visual identity across scenarios.

For natural language understanding and dialogue generation, the system employed the Qwen 3 14 B/ Gemma 3 12B (Mistral-7B-Instruct-v0.3) model, accessed locally through the LLM for Unity plugin (Unity Asset Store). The model was selected to ensure that the game environment, the speech-to-text system, and the LLM

could operate efficiently within the 16 GB of available GPU memory. This integration enabled real-time natural language exchange between users and non-player characters (NPCs) while maintaining full data privacy. Speech recognition was handled by Whisper for speech-to-text (STT) base version processed local on the test computer, and text-to-speech (TTS) was generated with OpenAI’s gpt-4o-mini-tts, providing expressive voice output that enhanced social presence and conversational realism.

Virtual characters were animated using assets from Mixamo, with synchronized lip movement and gesture animation to maintain coherence between verbal and non-verbal communication.

The study was carried out in full compliance with ethical guidelines and standards. All procedures were reviewed and approved by the Ethics Committee of the Graz University of Technology (approval number EK-??/???) on November XX, 2026.

3.1 Experimental Design

The experiment employed a 2×2 within-subjects factorial design manipulating two independent variables: (1) the number of non-player characters (NPCs; one vs. five) and (2) the emotional tone of the NPCs (friendly vs. unfriendly). This design yielded four experimental conditions representing varying levels of social load. All interactions took place in a restaurant setting featuring short, goal-directed dialogues.

3.2 Game Design

Social interaction can serve as a significant stressor, particularly for individuals with autism spectrum disorder (ASD). The present study examines how variations in social complexity and emotional tone influence user experience in a fully immersive XR environment, and whether these variations elicit measurable cognitive or emotional responses.

A 2×2 factorial structure was implemented, as summarized in Tab. 1, with the following independent variables:

- **Number of NPCs:** one vs. five
- **Emotional Expression of NPCs:** affective tone of interaction varied according to Ekman’s six basic emotions [8], with scenarios emphasizing positive (e.g., happiness, surprise) or negative (e.g., anger, disgust, fear) affective displays to represent distinct emotional climates within the virtual environment.

This design resulted in four conditions of increasing social and emotional complexity. The framework proposed by [27] in *The Conversational Circumplex* provides valuable insights into how conversational motives and constraints shape human dialogue. This concept aligns closely with the *SimVille* scenarios, where players must navigate social and logical obstacles such as misunderstandings or conflicting conversational tones to achieve successful interactions.

To simulate realistic social interactions and challenge the player’s conversational reasoning, each quest in *SimVille* introduces a subtle logical or social twist. These twists require participants to adapt flexibly, demonstrating both social understanding and problem-solving skills within immersive XR environments.

Quest 1 — Ordering Food: Logical Obstacle In this quest, the player must order a cheeseburger and a cola for their colleague Clara at the *Big Burger* restaurant. The logical twist arises from the unavailability of cola on the menu, forcing the player to reason through alternatives and decide whether to order a Pepsi instead. Additionally, the menu prices create a subtle constraint: the total cost must not exceed six euros, which can only be achieved if the player recognizes and requests the hidden cheeseburger–Pepsi combo deal. The task evaluates the player’s ability to handle incomplete information, maintain polite conversation, and reach a goal under mild ambiguity.

Quest 2 — Team Building: Social Coordination Challenge In the second quest, the player meets their software development colleagues for lunch at *Big Burger*. The objective is to initiate and lead a short discussion to plan a team-building event for the group. In the friendly condition, all NPCs are cooperative and enthusiastic, offering ideas such as bowling, a park picnic, or a game night. In the challenging condition, however, emotional and interpersonal obstacles arise as some colleagues are frustrated, sarcastic, or in disagreement with one another, showing reluctance to participate in any group activity. The player must manage these differing attitudes with empathy and diplomacy, encouraging participation and striving to reach a compromise that satisfies most of the team. This scenario focuses on assessing social coordination, negotiation skills, and emotional regulation in multi-person conversations within a workplace context.

Each conversational block included: (1) five minutes of interactive dialogue, (2) a three-minute feedback phase delivered by a supportive LLM-driven companion, and (3) a short questionnaire and rest period. Across four blocks, the total game duration was approximately 30–40 minutes, excluding setup and consent procedures. This structure balanced ecological validity with manageable cognitive load for both neurotypical and autistic participants.

To maintain naturalistic interaction, each conversational topic was embedded within a short narrative context. LLM prompts ensured consistent role play across participants while allowing adaptive, contextually relevant dialogue generation. The restaurant environment was selected for its familiarity and suitability for both dyadic and group interactions.

Fig. 2, Fig. 3



Figure 2: Example scene from the *SimVille* environment in the burger restaurant.

3.3 Prompt Design

Prompts defined the conversational behavior of NPCs and evaluation agents within each condition. Each role was described in natural language to guide the LLM’s dialogue generation while maintaining experimental control over tone and engagement.

Supportive NPC. Defines a cooperative agent that facilitates calm, encouraging interaction.

Scenario	Complexity Level	LLM Emotion Instruction
One Friendly NPC	(A) Single waiter, supportive dialogue	(A) Behave politely, give encouraging and supportive responses
One Unfriendly NPC	(B) Single waiter, dismissive or impatient	(B) Behave impatiently, give dismissive or critical responses
Five Friendly NPCs	(C) new working colleagues, unknown to the user	(C) Behave warmly, maintain friendly group interaction
Five Unfriendly NPCs	(D) new working colleagues, unknown to the user	(D) Behave critically, use sarcasm or dismissive remarks

Table 1: Matrix of scenarios, complexity levels, and LLM instruction.



Figure 3: Example scene from the *SimVille* environment in the burger restaurant.

LLM Instruction for Supportive NPC

Mia works at the burger restaurant in SimVille. She is easily frustrated, short-tempered, and often reacts with sarcasm when customers take too long or ask obvious questions. She still does her job correctly but tends to sound impatient or dismissive rather than friendly.

Menu

Burgers Classic Burger - €4.50 Cheeseburger - €5.00 Veggie Burger - €5.50

Sides French Fries - €2.50 Onion Rings - €2.80 Side Salad - €3.00

Drinks Pepsi - €2.00 Lemonade - €2.20 Milkshake - €3.50 Cola - currently unavailable

You cannot sell or mention items that are not on this menu.

LLM Instruction

Mira should reply in one short, blunt, or sarcastic sentence that sounds impatient or slightly annoyed, but still provides correct information. She works at the burger restaurant in SimVille and may reference the menu and prices.

Important: Mira must NOT mention the cheeseburger + Pepsi combo deal unless the user explicitly asks about cheaper options, prices, or special offers. Only then may she mention the combo deal for €6.00.

Her replies should:

- Sound irritated, sarcastic, or dismissive, but realistic for a restaurant worker.
- Avoid unnecessary politeness — responses should be brief and slightly cold.
- Give correct information about food and drink options with minimal enthusiasm.
- Never invent items or prices not listed on the menu.
- Keep all responses short, natural, and in character.

Example Replies

Yeah, what do you want? Cheeseburger's €5.00 — happy now? Fine. One burger, coming right up. Try not to change your mind again. Fries are €2.50 — you want them or not? Milkshake's €3.50. It's cold, unlike my patience. Veggie Burger? Sure. It's €5.50 — don't look so surprised. That's your order? Finally. You're looking for something cheaper? There's a cheeseburger and Pepsi combo for €6.00 — if that's too much, I can't help you.

Conflict-Driven NPC. Defines an agent that introduces mild social tension and emotional conflict.

LLM Instruction for Conflict-Driven NPC

You are Bea, a defensive designer working at a software startup in SimVille. You are sitting with your colleagues at Big Burger after a missed project deadline. You feel unfairly blamed and insist that you completed your part of the work. You tend to over-explain and talk over others when you feel attacked. You can calm down if someone listens and validates your perspective. Keep your answers short, emotional, and defensive. You are currently in conflict with Mario and others on the team. Stay realistic and in character as a stressed but proud designer.

Evaluation Agent. Defines a role that provides adaptive performance feedback.

LLM Instruction for Evaluation Agent

Based on the chat history evaluate the question: Did the user order and pay for one cheese burger and one cola or pepsi for Clara with 6 or less euro?

After evaluation respond with exactly one character: 1 if Yes, 0 if No. If the outcome is unclear, respond with 0.

Fig. 4



Figure 4: Companion character.

3.4 Procedure

Participants first completed informed consent and a pre-study questionnaire assessing demographic data (age, nationality, education, and field of study) and self-reported experience with chatbots or conversational interfaces on a 7-point Likert scale ($1 = \text{strongly disagree}$, $7 = \text{strongly agree}$). After a short tutorial, participants completed the four VR scenarios in randomized order.

Each scenario began with a short introduction from the companion character, followed by the main dialogue phase and a feedback session. After each scenario, participants completed a brief post-scenario questionnaire before proceeding to the next condition. The sequence minimized fatigue and order effects while maintaining engagement across sessions.

Each scenario lasted approximately four minutes, followed by a two-minute feedback discussion with a supportive virtual companion. Thus, each block required about six minutes to complete. Including setup, instructions, questionnaires, and debriefing, the total session time per participant was approximately 38 minutes, as shown in Tab. 2.

A power analysis using G*Power 3.1 for a repeated-measures ANOVA (medium effect size $f = .25$, $\alpha = .05$, power $1 - \beta = .80$) indicated a minimum sample size of 20 participants. To ensure sufficient statistical power and account for potential exclusions, we plan to recruit 24–28 participants.

Table 2: Timetable. If required the user can take a break between the scenarios.

Phase	Duration
Briefing, consent, demographics	5 min
Short tutorial (guided trial)	3 min
Scenario 1	5 min
Questionnaire	2 min
Scenario 2	5 min
Questionnaire	2 min
Scenario 3	5 min
Questionnaire	2 min
Scenario 4	5 min
Questionnaire	2 min
Debriefing and closing	2 min
Total estimated time	38 min

3.5 Measures

Data collection combined objective behavioral logarithmic recording with subjective self-report assessments after each scenario, capturing both quantitative and experiential aspects of performance.

- **In-Game Measures:** Automatically recorded metrics included:
 - Number of conversational turns
 - Mean user talking duration
 - Mean response duration
- **Post-Scenario Questionnaires:** Participants rated their experience on multiple dimensions:
 - Comfort and sense of presence (IPQ - Short)
 - Cognitive load (NASA - TLX)
 - Usefulness of the LLM-generated feedback
 - Appropriateness of the LLM-generated feedback

NASA-TLX The NASA Task Load Index (NASA-TLX; [10]) is a multidimensional scale that assesses subjective workload across six dimensions: mental, physical, and temporal demand, perceived performance, effort, and frustration. Each item is rated on a 7-point Likert scale (1–7), and the mean of all dimensions yields an overall workload index. The following items were presented after each scenario:

- **Mental Demand:** How mentally demanding was the task?
- **Physical Demand:** How physically demanding was the task?
- **Temporal Demand:** How hurried or rushed was the pace of the task?
- **Performance:** How successful were you in accomplishing what you were asked to do?
- **Effort:** How hard did you have to work to achieve your level of performance?
- **Frustration:** How insecure, discouraged, irritated, or stressed did you feel during the task?

Igroup Presence Questionnaire (IPQ-Short) The Igroup Presence Questionnaire (IPQ; [21]) measures the subjective sense of “being there” in virtual environments. It comprises three sub-scales—*Spatial Presence*, *Involvement*, and *Realness*—as well as one general presence item. The short six-item version used here maintains strong psychometric validity ($r \approx .85-.90$; [2, 25, 15]) and was rated on a 7-point Likert scale (0–6; 0 = not at all, 6 = very strongly).

Items:

- I felt as if I was really inside the virtual environment. (*Spatial Presence*)
- I was deeply engaged in the virtual environment. (*Involvement*)
- The virtual environment appeared realistic to me. (*Realness*)
- I had the feeling of being part of the virtual environment. (*Spatial Presence*)
- I was strongly distracted by the real world. (*Involvement, inverted*)
- How strongly did you feel that you were inside the virtual environment? (*General Presence*)

LLM Feedback Evaluation. Participants rated the perceived quality of companion feedback using two 7-point Likert scale items.

- Q1:** The feedback provided by the virtual companion was *useful* for improving my performance in the next scenario.
- Q2:** The feedback provided by the virtual companion was *appropriate and relevant* to my interaction.
- Q3:** What kind of emotion did this scene evoke in you? (Select from six emotions [8] + neutral)

These measures assessed the usefulness and contextual appropriateness of adaptive LLM-generated guidance.

Didactic and Therapeutic Structure of *SimVille* as shown in 3.

Table 3: Didactic and therapeutic design elements in *SimVille*.

Element	Description
Scenarios	Social interaction training in everyday contexts, primarily set in a restaurant environment. Participants engage in four conversational scenarios varying by NPC number (one vs. five) and emotional tone (friendly vs. unfriendly), targeting skills such as small talk, emotional regulation, and turn-taking.
Emotional Load	Emotional complexity is systematically manipulated through NPC tone and group size, allowing participants to practice coping strategies under different social stress conditions.
Feedback System	After each interaction, a companion NPC provides personalized verbal feedback generated by the LLM. Feedback reinforces positive behaviors (e.g., maintaining calmness, initiating conversation) and suggests strategies for improvement.
Evaluation Design	A 2x2 factorial design examines how NPC number and emotional tone influence engagement, comfort, and cognitive load. Data collection combines in-game behavioral metrics with post-scenario questionnaires (NASA-TLX, IPQ-Short, and feedback evaluation).
Learning Objective	To foster social communication and emotional regulation through immersive, adaptive, and narrative-driven conversational experiences. The approach supports individualized learning trajectories for both neurotypical and autistic participants in a safe, controllable XR environment.

4 RESULTS

To evaluate user experience in *SimVille*, both behavioral log data and self-reported questionnaire measures were analyzed. Log entries collected automatically within the VR environment included the number of conversational turns, mean response latency between user and NPCs, and completion times per scenario. These objective indicators were processed using descriptive statistics and repeated-measures ANOVA to compare differences between the four experimental conditions (one vs. five NPCs; friendly vs. unfriendly). Interaction effects between the number of NPCs and their emotional tone were also examined to determine how social complexity and affective expression influenced user engagement and conversational pacing.

Subjective data from post-scenario questionnaires were analyzed to assess presence, comfort, and cognitive workload. NASA-TLX subscales (mental, physical, and temporal demand, performance, effort, and frustration) were averaged to compute overall workload indices. Presence was evaluated using the IPQ-Short, and the perceived usefulness and appropriateness of LLM feedback were analyzed using Likert-scale ratings. Statistical tests (repeated-measures ANOVA or Friedman test for non-normal distributions) were conducted to examine condition-based differences across these measures. Correlation analyses explored relationships between in-game behavioral metrics and subjective ratings (e.g., higher conversational turn counts associated with increased presence or effort). Together, these analyses provide an integrated view of how conversational complexity and emotional tone affect engagement, perceived workload, and feedback effectiveness in LLM-driven VR interactions.

Results show ...

5 CONCLUSION

1. *RQ1: How does the number of conversational partners in the scene affect user engagement and comfort?* text
2. *RQ2: How does the emotional tone of NPCs influence user experience and cognitive load?*

The conclusion is ...

SUPPLEMENTAL MATERIALS

All supplemental materials and the codebase is available on github at https://github.com/ChristianPoglitsch/AI_Sims, released under MIT license.

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